

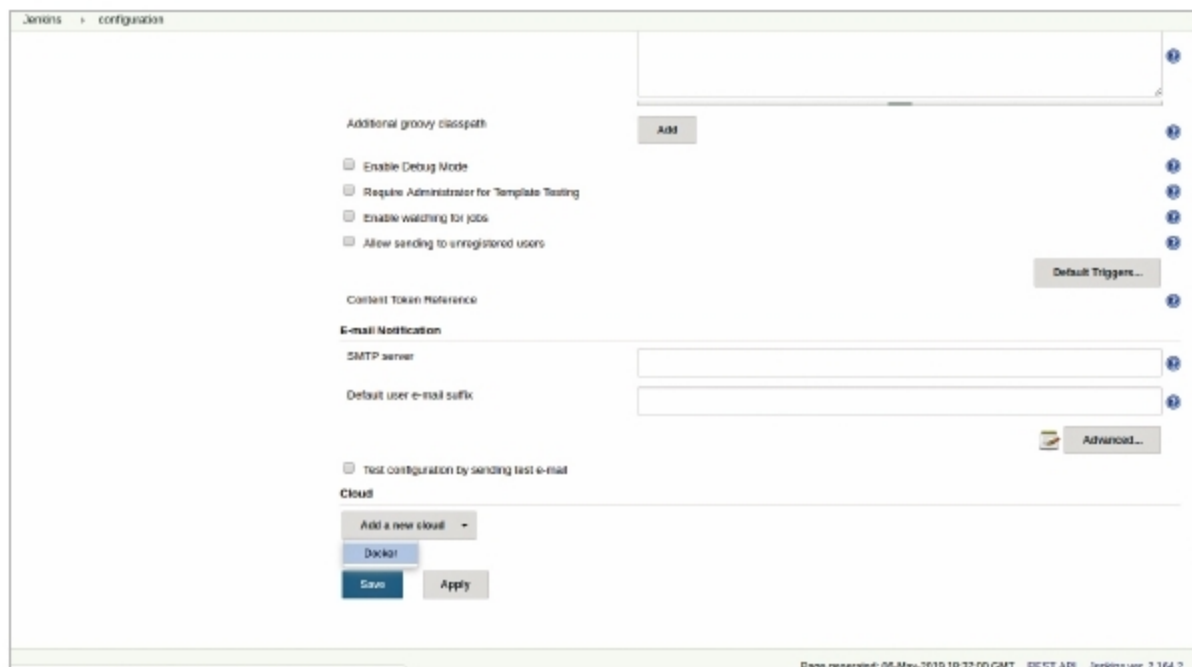


Figure 7: Docker setup in Jenkins

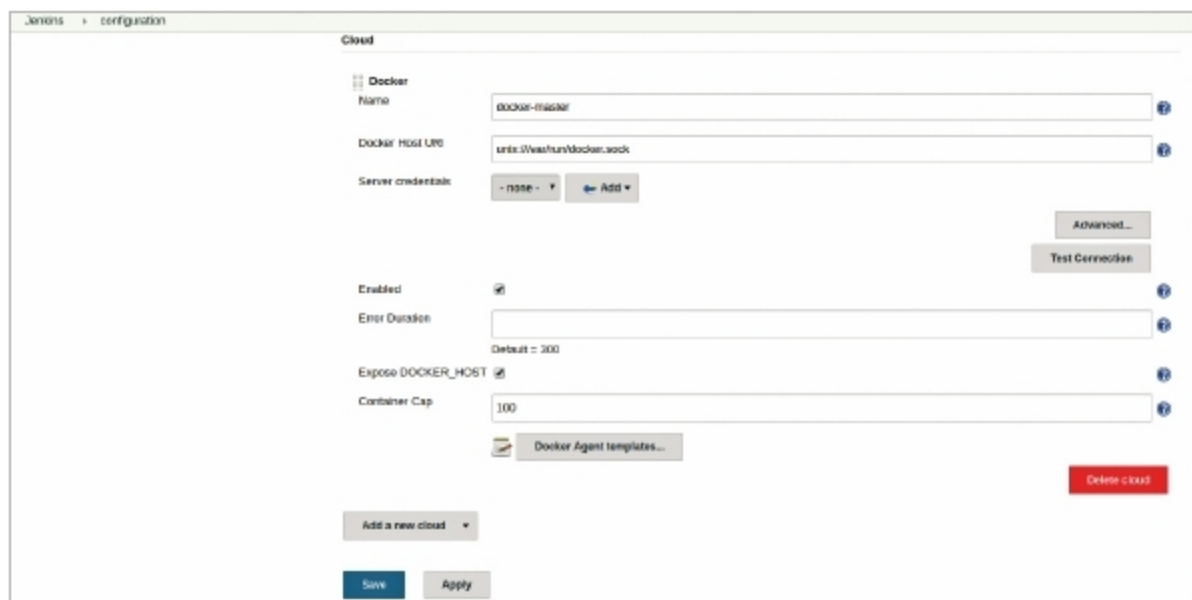


Figure 8: Docker configuration

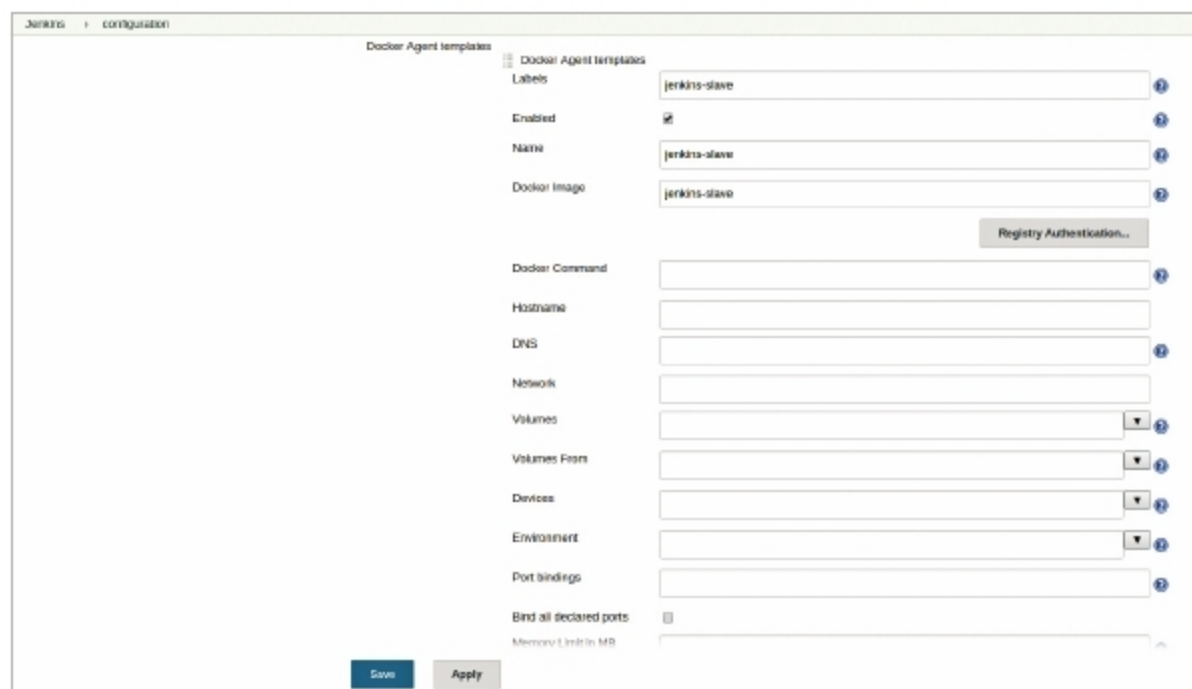


Figure 9: Agent template configuration

Configuring a Jenkins master on Docker: Before we spin up Jenkins on Docker, we should have Docker installed on our machines.

Pre-requisites:

- Docker can be installed on Mac, Linux and Windows OSs.
- If the machine is running on Windows, then it should be either

Windows 10 Pro or the Enterprise version, because the other versions of Windows lack Hyper-V support.

Once Docker has been set up on the machine, we can execute Docker commands. Let's run some Docker commands and check the version, and see if any containers exist.

The commands are:

Version - `docker -version`
Containers - `docker ps -a`

In Figure 3, the version has been displayed and there are no containers running in the Docker system.

Note: Jenkins officially recommends the `jenkinsci/blueocean` image to run the Jenkins server.

In order to spin up the container, this image needs to be downloaded first. The image can either be downloaded using the `docker pull <image_name>` command, or this can be done directly using the `docker run` command and mentioning the image name. This will check the Docker images, and if the image is not available, it will be downloaded.

As I have already downloaded the image, I have directly issued the `docker run` command, as shown in Figure 4. When a Docker image is run successfully, the container is assigned a container ID, which is given as a return value on executing the `docker run` command.

The command executed is:

```
docker run -u root -d 8080:8080 -p
50000:50000 -v /home/ajay/Desktop/
jenkins_home:/var/Jenkins_home/ -v /
var/run/docker.sock:/var/run/docker.
sock --name Jenkins --restart always
jenkinsci/blueocean
```

This is a complete one-line command. When `docker run` is fired, the minimum mandatory argument is the name of the image.

Table 4 lists Docker commands and gives a brief description of each.

The Docker container can be checked using the `docker ps -a` command. Now, when we check the